Inventory Management System Project Report

Python, Java, and Data Analytics Implementation

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WEEK 4: May 25, 2026 – May 31, 2026

Project Overview

The Inventory Management System project focuses on designing and implementing an efficient software solution to manage stock levels, product tracking, forecasting, and reporting. The system integrates analytics tools to improve inventory decisions and operational efficiency.

Objectives

- Develop real database-driven CRUD APIs for products and suppliers using Spring Boot and SQLite.
- Implement JSON-based transaction and low-stock alert APIs for efficient data exchange.
- Connect backend APIs with the frontend dashboard for real-time inventory management.
- Enhance system functionality through forms, testing, and documentation updates.

Roles and Responsibilities

- Developed and integrated Product, Supplier, and Transaction CRUD APIs with SQLite.
- Implemented JSON responses and low-stock alert endpoints for backend services.
- Connected frontend dashboard components with backend APIs and created input forms.
- Performed API testing using Postman and maintained project documentation and GitHub updates.

Tools and Technologies Used

Python, Java, Spring Boot, SQLite, SQL, Pandas, Matplotlib, GitHub, VS Code, Postman

Key Outcomes

- Successfully implemented database-connected CRUD operations and JSON APIs.
- Established frontend-backend integration for inventory monitoring and management.
- Improved practical knowledge of full-stack development, API design, and database integration.

Conclusion

Week 4 focused on transforming the system into a fully functional database-driven application with real API connectivity. The completed enhancements significantly improved system usability, scalability, and readiness for advanced inventory management features.

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